Study guide: PFP II

AuthorityEffect: None. Start with the fictional Apartment 50 preface. This guide prepares the 14-problem quiz; complete an attempt before opening the separate worked instructor key. All required readings and checks are included in this deposit.

What you need and what you should be able to do

Prerequisites are addition modulo two, finite graph walks, matrix multiplication, exact fractions, a spanning-tree argument, factorials, and the power series for the exponential. For the Fock example, use the manuscript's explicit occupation and hopping rules; no prior many-body simulation experience is required. Review those rules before attempting the integration and length bounds.

By the end, produce and independently check both witness types; derive the matrix adapter; distinguish a complete classifier from an unfinished enumeration; replay a contiguous receipt chain; and explain exactly which pairing operation the Fock example excludes.

Notation and the objects being checked

Object	Meaning	Typical mistake
$c_e \in \mathbb{Z}_2$	An edge's endpoint constraint	Treating XOR as rational addition
α_v	A vertex bit; sign $(-1)^{\alpha_v}$	Assuming the chosen root removes candidates from the implementation
Odd closed walk	Indexed traversals, closure, parity one	Checking parity without adjacency or closure
Exact matrix	Integers or `Fraction` values in a fixed basis	Silently rounding a floating input
Presentation key	Full declared context for retrieval and replay	Matching only the Hamiltonian digest
Receipt frontier $[s, t)$	The suffix outside the recorded checks	Treating an unfinished suffix as a negative proof
Integrated rooted path	A state sequence whose jump times were integrated out	Equating finite state count with finite sequence count

An arrow records a traversal in a walk. A graph edge does not attach a two-dimensional cell. A bracket-wire encoding, if used, needs its own declared interpretation and parsing rules.

Route through Levels 0-7

Level	Read in the manuscript	Practice	Completion check
0	Sections 1-2	Q1	State the carrier and exact move grammar
1	Section 2	Q2-Q3	Check a gauge and construct indexed obstructions
2	Section 2	Q4-Q5	Separate coefficients and prove both directions
3	Sections 1 and 3	Q6-Q7	Compute DHD and prove exponential positivity
4	Section 4	Q8-Q10	Account for cursor, batch, key, and prior coverage
5	Section 5	Q11	Check an actual pairing map, not just its sum
6	Section 6	Q12-Q13	Derive the target and cover every length class
7	Sections 7-8 and the claim ledger	Q14	Match an exact claim to its evidence and limit

1. Local equations and the global witness

Practice first on a tree: root one vertex at zero and propagate its bits. There is no cycle constraint to violate. Add a closing edge and compare the required endpoint bit with the propagated bit. A disagreement supplies an odd closed walk through the tree.

For a gauge, check every edge equation, including loops and repeated endpoints. For an obstruction, check every edge ID and endpoint, continuity between steps, final closure, and the XOR of its bits. Reversing a traversal does not change a binary edge bit. Give each parallel edge its own identity.

Then change coefficients. Rational reports on an oriented cycle require a telescoping sum of zero. Modulo-two parity can vanish while that rational sum does not. Keep the coefficient system and allowed report repairs in the statement.

2. The matrix bridge

The adapter assigns a bit one to a positive nonzero off-diagonal entry and zero to a negative one. This bit is the constraint needed to make the transformed matrix entry negative; the graph sign $(-1)^{c_e}$ follows that convention. Zero entries impose no equation.

Compute an entry of DHD as $D_{ii}H_{ij}D_{jj}$ before applying a general rule. A successful gauge proves stoquasticity under those diagonal signs in the declared basis. Other basis changes require a different grammar.

For exponential positivity, shift the diagonal so $B = aI - H'$ is entrywise nonnegative. Powers of B remain nonnegative, and the convergent power series gives the result for nonnegative β. This establishes matrix-entry positivity; sampling cost and observable-estimator variance require additional analysis.

3. Complete classification and incomplete search

The graph classifier is linear in the finite input size. Exhaustive enumeration is retained to exercise a protocol for future grammars; its worst-case candidate count is exponential.

Search status	Evidence required	What follows
`GAUGE_VERIFIED`	A gauge passes all equations	That gauge works in this grammar
`UNKNOWN`	The recorded segment has no valid gauge and a suffix remains	Continue only with a verified prior prefix
`RESTRICTED_OBSTRUCTION`	Replayed exhaustive coverage and a checked odd walk	No vertex gauge exists for this graph

The direct graph classifier can supply an odd walk without exhaustive enumeration. The search function issues its restricted terminal status only after its own finite grammar is exhausted.

For the three-vertex example the implementation has eight candidates, indexed zero through seven, even when a proof fixes a root. Track `start_index`, ordered checks, `next_index`, and `stop_index`. The cursor advances by the full consumed batch, not necessarily to one past the first valid candidate.

A digest binds bytes; replay checks their asserted equations. Continuation requires a contiguous chain of UNKNOWN segments beginning at zero, with the

same graph and presentation digests. Changing the basis, grammar, estimator, adapter, or another key field changes the declared context. A retrieved bit vector must be checked again even after a full-key match.

4. Cancellation and the infinite family of lengths

A pairing is a map on a closed paired subset. Check involution, absence of fixed points, opposite signs, and equal magnitudes. Separate unpaired survivors from that subset. A nonnegative total does not by itself supply a pairing.

For the Fock example, keep the six-mode order fixed. Count occupied modes strictly between a hop's endpoints for its sign and doubly occupied sites for its energy. A constant-energy three-hop sequence has magnitude $e^{-\beta E}\beta^3/3!$.

The proof has two different parts: exact enumeration establishes the special facts for length three, while inequalities cover every other length. For $n \geq 4$, use

$$\frac{\beta^{n+1}/(n+1)!}{\beta^n/n!} = \frac{\beta}{n+1} < 1 \quad (\beta = 1/4).$$

This controls an individual sequence. Counting and summing many sequences is a separate task.

For $x \in [0, 1]$, odd truncations of the alternating series for e^{-x} are lower bounds and even truncations are upper bounds. Compare rational enclosures instead of rounded decimals. The target lies above every long-path upper bound and below every relevant positive short-path lower bound, excluding an equal-magnitude partner.

5. Replay the package and describe the evidence

From the package root, use the README commands. On Windows use `py -3.12`; elsewhere substitute your Python 3.12 interpreter. The runtime checkers use the standard library. Tests require pytest; notebook replay additionally requires nbclient, nbformat, and ipykernel.

Replay	Supports	Does not establish
`code/tests/test_sign_graph.py`	Input and signed-witness checks	A new graph theorem
`code/tests/test_manuscript_examples.py`	Displayed matrix and 2-then-1 search trace	General physical adapter validity
`code/tests/test_frontier_search.py`	Coverage, ordering, key, and snapshot controls	Sampling speedup
`code/exact_fock_gap.py` and its test file	Exact counts and rational gap checks	Exclusion of every representation

Replay	Supports	Does not establish
Native-adapter tests	Adapter behavior; fallback if no executable exists	A native build when binary discovery skips
Fresh notebook replay	The deposited notebook executes under the recorded environment	Independent experimental data
`verify_deposit.py` and release ZIP check	File census, hashes, archive identity	Truth of a theorem

The claim ledger maps the paper's statements to proofs and named checks. Consult the current dated review receipt for actual execution, versions, and skips; historical counts keep their original scope.

Finish with a defense

Complete the quiz, mark it against the separate key, and write down the reason for each error: wrong carrier, wrong coefficients, invalid witness, missing coverage, or overextended claim. Rework the associated example before the oral defense. The score and mastery target are teaching conventions; no learner study or automatic authority is claimed.